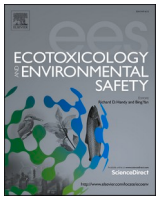
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Microbial community function and methylmercury production in oxygen-limited paddy soil

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ABSTRACT

Methylmercury is a neurotoxic compound that can enter rice fields through rainfall or irrigation with contaminated wastewater, and then contaminate the human food chain through the consumption of rice. Flooded paddy soil has a porous structure that facilitates air exchange with the atmosphere, but the presence of trace amounts of oxygen in flooded rice field soil and its impact on microbial-mediated formation of methylmercury is still unclear. We compared the microbial communities and their functions in oxygen-depleted and oxygen-limited paddy soil. We discovered that oxygen-limited paddy soil had higher methylmercury concentration, which was strongly correlated with soil properties and methylation potential. Compared with oxygen-depleted soil, oxygen-limited soil altered the microbial composition based on *16 S rRNA* sequences, but not based on *hgcA* sequences. Moreover, oxygen-limited soil enhanced microbial activity significantly, increasing the abundance of more than half of the KEGG pathways, especially the metabolic pathways that might be involved in methylation. Our study unveils how microbial communities influence methylmercury formation in oxygen-limited paddy soil.

Environmental implications: This study examined how low oxygen input affects microbial-induced MeHg formation in anaerobic paddy soil. We found that oxygen-limited soil produced more MeHg than oxygen-depleted soil. Oxygen input altered the microbial community structure of *16 S rRNA* sequencing in anaerobic paddy soil, but had little impact on the *hgcA* sequencing community structure. Microbial activity and metabolic functions related to MeHg formation were also higher in oxygen-limited paddy soil. We suggest that oxygen may not be a limiting factor for Hg methylators, and that insufficient oxygen input in flooded paddy soil increases the risk of human exposure to MeHg from rice consumption.

1. Introduction

Mercury (Hg) is a global pollutant that can travel long distances and contaminate the atmosphere, soil, water, and other environmental media (Dai et al., 2021; Li et al., 2022). Its toxicity depends on its chemical form, with methylmercury (MeHg) being the most dangerous due to its high neurotoxicity and its tendency to accumulate in the food chain (Beckers and Rinklebe, 2017; Gustin et al., 2020; Mortazavi et al., 2016). Rice (*Oryza sativa* L.) and rice-derived foods are a staple food for more than half of the world's population, and the rising levels of MeHg in rice have raised global concerns (Feng et al., 2021; Guo et al., 2021). In China, rice consumption has been identified as a major source of exposure to MeHg from Hg mining (Qiu et al., 2008). To reduce the risk of Hg exposure from rice consumption, it is essential to minimize the amount of Hg released into the environment from Hg mining. China has committed to ban Hg mining by 2032 as part of this effort. Although global atmospheric Hg concentrations have been declining, the discharge of Hg-contaning wastewater (e.g., acetaldehyde plants, chlor-alkali plants, and even domestic wastewater treatment plants) still contribute to increased Hg content in local soil. As a result, water should be properly treated before being used for irrigation (Panagopoulos, 2022; Panagopoulos and Giannika, 2022). In situations where wastewater treatment capacity cannot be achieved in the short term or where there is a historical legacy of Hg in the environment, research is needed to identify appropriate field management measures, such as water management (Rothenberg et al., 2016), to reduce or prevent the

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formation of MeHg.

Field investigations have demonstrated that the concentration of MeHg in soil or sediment is low when it remains in an oxidized or anaerobic state for a long time. However, at locations where the transition between oxidized and anaerobic states occurs frequently, such as water level fluctuations in the Three Gorges Reservoir (Xiang et al., 2018) and stump harvest in boreal forest (Eklöf et al., 2018), the concentration of MeHg is significantly higher than that of the surrounding environment. A growing body of evidence suggests that the methylation of Hg is influenced by constantly changing environmental conditions, which strongly affect the occurrence and activity of methylating microorganisms, as well as the bioavailability and methylation potential of inorganic mercury species (W. Tang et al., 2020). Flooded rice paddy soils are a typical changing environment and a hotspot for MeHg formation (J. Liu et al., 2019; Wang et al., 2019). Research has shown that MeHg in rice mainly originates from MeHg produced in paddy soil (J. Liu et al., 2019). The formation and uptake of MeHg are influenced by various environmental factors, such as soil redox potential, pH, organic matter, temperature, and mineral composition (Feng et al., 2021; Luo et al., 2023; Z. Tang et al., 2020; Wang et al., 2021; Zhao et al., 2020). Furthermore, it has been found that the MeHg accumulated in the food chain is not directly emitted by human activities, but is generated by microbial action in natural processes (Wang et al., 2021). Microbial-mediated Hg methylation is a crucial factor in regulating MeHg production and mobility. Currently, we cannot control MeHg pollution from the source, and the key lies in our insufficient understanding of the mechanism of MeHg generation in the environment (Branfireun et al., 2020; Luo et al., 2023; W. Tang et al., 2020).

The anaerobic environment of paddy fields provides ideal conditions for Hg methylation (Beckers and Rinklebe, 2017; Bowman et al., 2020), but it is unclear whether this anaerobic environment is the necessary factor responsible for the high methylation of paddy soil. The flooded paddy soil creates an anaerobic environment, which does not mean that no oxygen enters the soil, but because the oxygen consumption rate exceeds diffusion. Therefore, anaerobic rice fields can be divided into at least two types, oxygen-depleted paddy soil and oxygen-limited paddy soil (Feng et al., 2022; Mangal et al., 2019; Wang et al., 2023; Xu et al., 2019). As anaerobic microorganisms act differently than facultative aerobic ones to produce metabolites in the soil, studies of their communities may introduce new approaches to decrease Hg methylation (Sepehri and Sarrafzadeh, 2018). However, previous studies did not pay much attention to this difference, which may explain why there were many contradictory results when studying methylation in paddy fields. For example, methanogenesis act as methylating bacteria in rhizosphere soil and as demethylating bacteria in the surface soil (Wu et al., 2020). Yet, this changing oxygen condition has not been well considered as an important factor for MeHg formation in flooded rice fields.

Oxygen can enter paddy soil through diffusion, irrigation, biological disturbance, and plant root transport, thus affecting the production of MeHg (Lin et al., 2021; Malcolm et al., 2010). Intense gas exchange and oxygen consumption can influence Hg migration and transformation, as well as the related health risks of Hg (Malcolm et al., 2010). Previously Hg methylation was considered an anaerobic process of anaerobic microorganisms, including sulfate-reducing and fermenting bacteria, and the putative Hg methylators were strictly anaerobic bacteria or archaea (Regnell et al., 2019). However, recent studies using genetic sequencing have revealed a wider variety of potential methylators (Jones et al., 2020; Liu et al., 2022; Millera Ferriz et al., 2021; Xu et al., 2021; Yoshimura et al., 2020; Zhou et al., 2020), even aerobic bacteria (Cao et al., 2021; Feng et al., 2022). Oxygen has multiple effects on Hg-methylating microorganisms and non-Hg-methylating microbes, leading to the uncertainty of oxygen prediction of MeHg risk. A small amount of oxygen input may alter microbial-mediated Hg methylation. On one hand, oxygen input may harm anaerobic methylating bacteria, thus reducing MeHg production (Gilmour et al., 2018; Lu et al., 2016). On the other hand, oxygen input can enhance the growth and activity of both facultative Hg-methylating bacteria and non-Hg-methylating bacteria in the soil, thus promoting MeHg production (Liu et al., 2018; Y. Liu et al., 2019). Oxygen input can also increase the availability of nutrients in the soil, which can further stimulate the growth and activity of microbial community (Ma et al., 2018; Xiang et al., 2018). To better understand the effects of oxygen-limited anaerobic environment on MeHg production, more research is needed to improve MeHg risk prediction.

In this study, we test the hypothesis that oxygen input has multiple effects on MeHg production in paddy soil. We simulate Hg-contaminated wastewater irrigation by adding Hg-containing overlying water to previously uncontaminated soil. We control the ventilation conditions to simulate soil with depleted oxygen and limited oxygen. Subsequently, we measured changes in Hg mobility, microbial community and Hg methylators community to interpret the observed changes in MeHg formation. The results of this study will help reduce the uncertainty in predicting MeHg production and related risks in anaerobic paddy soils that are often influenced by low oxygen input from natural or human sources.

2. Materials and methods

2.1. Study site

The soil used in the experiment was carefully selected to ensure that it was not Hg-contaminated. The soil was collected from a rice paddy field in Changsha, China, and had a total Hg (THg) concentration of 0.062 mg/kg, which is lower than the average background level of THg (0.10 mg/kg, Chinese National Standard Agency, 2017). The soil was air-dried, mashed, and passed through a 2 mm sieve before the incubation experiments.

2.2. Experimental design

The soil resuspension microcosms were designed to simulate flooding conditions in paddy fields and to study the effects of Hg pollution on the soil. The microcosms were filled with 50 g of air-dried soil and 400 mL of water at a mass ratio of 1:8. 0.15 mL 1 g/mL of dissolve mercury nitrate was added to each microcosm to simulate moderate Hg pollution of farmland soil, resulting in a soil THg concentration of 3 mg/kg. To eliminate any C-limitation effect, 10 g of sodium lactate was added to each microcosm as a carbon source for microorganisms.

Since Hg concentrations in various human-influenced contaminated soils vary in the range from 0.04 to >100 mg/kg, the Hg concentrations chosen in previous experimental studies are not uniform. We chose a soil Hg concentration of 3 mg/kg, which represents both mercury-contaminated agricultural soils and does not harm the microbial community (Liu et al., 2014). We investigated Hg methylation in the paddy soils via a short-term (24 h) incubation experiment, because the methylation of mercury starts within 24 h after deposition (Chiasson--Gould et al., 2014; Gascón Díez et al., 2016), while the demethylation process has a delayed onset (Zhou et al., 2020).

The ventilation was designed to simulate a range of aeration conditions. The treatments consisted of (1) 24 h purging with nitrogen gas (N24), (2) 12 h purging with synthetic air (20.9 % O_2 + 79.1% N_2) followed by 12 h purging with nitrogen gas (N12), (3) 12 h purging with nitrogen followed by 12 h purging with synthetic air (O12), and (4) 24 h purging with synthetic air (O24). N24 and N12 indicate oxygen-depleted soil, while O24 and O12 indicate oxygen-limited soil. The gas flow rate was maintained at 500 mL/min for 24 h. Each treatment was replicated three times, resulting in 12 soil samples. The soil resuspension microcosms were incubated in the dark at room temperature (28 ± 1 °C) on a magnetic mixer at 300 rpm. This incubation period was chosen as it allowed MeHg production to increase, but prevented MeHg loss by demethylation.

2.3. Chemical analysis

All plastic containers used for Hg measurements were immersed in 4% HNO_3 (v/v) for at least 24 h and rinsed with ultrapure water before sampling. THg was measured according to GB/T 22105.1–2008. Briefly, approximately 1 g of freeze-dried soil samples were weighed and digested in aqua regia with an HCl to HNO_3 ratio of 1:1 (v/v) for 2 h at 90 ℃, and then diluted to 50 mL with Milli-Q water. The digested solutions were reduced by $NaBH_4$ to atomic Hg and imported into an atomic fluorescence spectrometry with a carrier gas (argon). MeHg was measured according to HJ 977–2018. Briefly, approximately 1 g of freeze-dried soil samples were weighed and digested using 15.0 mL of 25% KOH/CH_3OH for 3 h at 60 ℃. The soil solution was mixed with a vortex machine, centrifuged, and the supernatant was transferred to a new tube. The filtered supernatant samples were then ethylated into methylethylmercury, which was purged onto a Tenax trap in a stream of N_2. The concentration of the MeHg was determined by CVAFS (Brooks Rand Model III, Brooks Rand Labs, USA). The geochemical parameters ORP, pH, and dissolved oxygen (DO) in the soil solution were measured in situ using a multi-parameter water quality meter (Leici, China). Ferric was measured according to the phenanthroline spectrophotometric method. Sulfate was determined by turbidimetry, a method of measuring sulfate ions by measuring the amount of light scattered by a sample solution.

2.4. Microbial community composition of paddy soil samples

We employed the MoBio PowerSoil DNA Isolation Kit (MoBio Laboratories, Carlsbad, CA, USA) to extract total genomic DNA from soil samples following the manufacturer's instructions. The hypervariable v4 region of the *16 S rRNA* gene was amplified (GeneAmp 9700, ABI, USA) using universal bacterial and archaeal primers 515 F (5′-GTGCCAGCMGCCGCGG-3′) and 806 R (5′-GGACTACHVGGGTWTCTAAT-3′) (Liu et al., 2018). Subsequently, the purified PCR products were visualized for the correct size by 2% agarose gel electrophoresis and then extracted using the AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA, USA). Purified Amplicons were sequenced in a paired-end format using the Illumine MiSeq PE300 platform by Majorbio BioPharm Technology Co, Ltd (Shanghai, China). The high-throughput sequencing data were demultiplexed and quality-filtered according to the QIIME 2 pipeline. Operational taxonomic units (OTUs) were clustered with a 97% similarity cutoff using UPARSE (version 7.1) and chimeric sequences were identified and removed using UCHIME. The taxonomy of each *16 S rRNA* gene sequence was analyzed by TaxAss against the nucleotide sequence database using a confidence threshold of 70%.

2.5. Hg methylators community composition: hgcA

For microcosm soil samples, *hgcA* genes were amplified using the primers *hgcA*_261F (5′-CGGCATCAAYGTCTGGTGYGC-3′) and *hgcA*_912R (5′-GGTGTAGGGGGTGCAGCCSGTRWARKT-3′; GeneAmp 9700, ABI, USA) (Schaefer et al., 2014). The list of *hgcAB* organisms ('ORNL-Hg-methylator-list-updated-10162018') was published as supplementary information in (Christensen et al., 2019). The gene sequence *hgcAB* was amplified by the method described by (Gionfriddo et al., 2020). Briefly, Nucleotide *hgcA* sequences were aligned (MUSCLE), trimmed (201 bp), translated into amino acids using Geneious version 10.2, and compiled into a hmm-model using 'hmmbuild'. A maximum likelihood (mL) *hgcA* distribution was estimated using the GAMMA model of rate heterogeneity and mL estimate of alpha-parameter in RAxML with 150 bootstrap replicates. Taxonomic classifications can be assigned with the lowest common ancestor (LCA) of the subtree where the sequence was placed using 'guppy classify' with a classification cutoff of 90%.

2.6. Data analysis

A one-way ANOVA was conducted to assess the significance of differences among the treatments. LSD test was used to determine the mean differences among treatments, with P-value of less than 0.05. The alpha diversity indexes including the Chao, Sobs, Shannon, Simpson were computed using the 'mothur' package. The Kruskal-Wallis test was used to compare the differences between the various samples, followed by the Turkey-Kramer posthoc tests. A redundancy analysis (RDA) was conducted in R version 4.2.2 using the 'vegan' package to determine the relative importance of microorganisms and selected soil factors in governing Hg-methylation. The results of the analysis revealed the relative importance of Hg-methylating microorganisms and soil factors in governing Hg-methylation in soil. The differences in beta diversity of bacterial communities in different samples were estimated using Principal Coordinates Analysis (PCoA) based on Bray-Curtis and Weighted Unifrac distances, and the significance was assessed with the PERMANOVA test using the 'vegan' packages in R version 4.2.2. The functional analysis of bacterial communities was conducted using Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt2), based on the taxonomic annotation of *16 S rRNA* gene sequences. Functional inferences were derived from the Kyoto Encyclopedia of Genes and Genomes (KEGG). Statistical tests revealed significant differences with P-values of less than 0.05.

3. Results

3.1. THg, MeHg, and soil properties in different oxygen treatments

To examine how low oxygen input affects MeHg production under anaerobic soil, we kept the DO concentration of all samples below 0.01 ppm. The results of the experiment indicate that the majority of the Hg added to the soil was stored in the soil, with only a small amount of dissolved THg and MeHg present in the water. The mean values for dissolved THg and MeHg were 2.96 μg/L and 1.50 ng/L, respectively. The MeHg concentration in the soil ranged from 83.2 to 173.2 μg/kg, with a mean value of 121.4 μg/kg. The O24 group had the highest concentration of MeHg of 132.6 μg/kg, and the N24 group had the much lower concentration of MeHg of 74.9 μg/kg, while the half-stress groups (N12, O12) showed a slight change in MeHg concentration, though without any significant difference (Fig. 1A). The THg concentration in the soil ranged from 2.37 to 3.66 mg/kg, with a mean value of 2.83 mg/kg. The relative changes in each treatment group were similar to those of MeHg, with the O24 group also showing a significant increase over the other three groups, reaching a maximum value of 3.34 mg/kg. the O12 group had the lowest mean value, but this should be attributed to the fact that a minimum value in O12 group was measured (Fig. 1B). The methylation potential (the ratio of MeHg and THg, %MeHg) in the soil ranged from 2.92% to 4.27%, with a mean value of 3.49%. However, when comparing the %MeHg among the treatments, the difference between the O24 and N24 group was not statistically significant, while the O12 and N12 group showed a statistical difference, indicating a greater range of fluctuations in the effect of short periods of oxygen flux on MeHg production (Fig. 1C).

ANOVA tests showed that ventilation conditions had significant effects on pH, ORP and electron acceptors sulfate and ferric in paddy soils, with significant differences between the different treatments. However, the time of oxygen introduction did not vary linearly with soil properties, as the highest pH and ORP values were observed in the O12 group, and much lower pH and ORP values were observed in the N12 group, indicating that transient changes in oxygen conditions have a greater effect on soil properties than stable oxidizing conditions (Fig. 2A, B). In addition, microorganisms can transform Hg into MeHg. Sulfate and ferric are key electron acceptors for the microbial methylation process. The result indicates that N24 group has the highest concentrations of sulfate and ferric, 27.3 mg/kg and 0.926 mg/kg respectively, which are

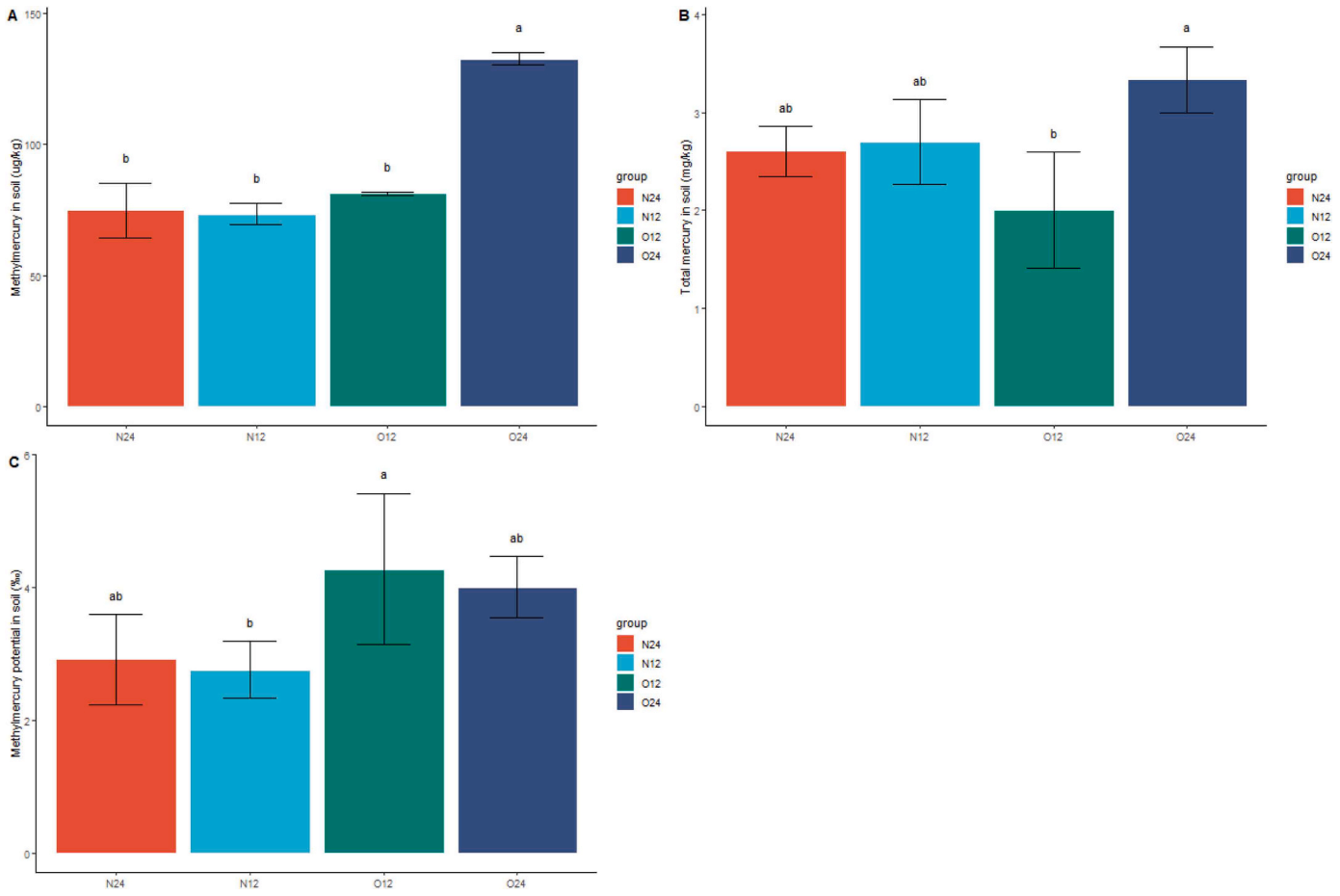


Fig. 1. Effect of nitrogn and/or oxygen amendment on mercury in soil. (A) methylmercury concentration, (B) total mercury concentration, (C) methylmercury potential. The lowercase letters and corresponding error bars indicate significant differences ($p < 0.05$). Data are shown as the mean ± standard deviation.

close to the soil background values. The concentrations decrease in N12 group, while O12 and O24 groups use up more electron acceptors. This implies that the consumption rate of these electron receptors increases after exposure to oxygen, which may be related to the activity of sulfate-reducing bacteria and iron-reducing bacteria. The availability of the electron acceptors could affect the bioavailability of Hg and thus the capacity of microbial communities to perform Hg methylation. It is noteworthy that the concentration of ferric was lower in O12 group than in O24 group, which agrees with the measurements of pH and ORP (Fig. 2C, D).

3.2. 16S rRNA sequencing community composition of the paddy soil

The composition of bacteria and archaea in 12 paddy soil samples was determined by *16 S rRNA* high-throughput sequencing. After optimization, a total of 1125789 sequences with 288203147 bases and an average sequence length of 256 bp were obtained. Based on the minimum sequence number (67589), taxonomic analysis revealed that there were 2 domains, 2 kingdoms, 31 phyla, 84 classes, 166 orders, 250 families, 423 genera, 660 species, and 1427 OTUs. Notably, the sequences belonging to Archaea accounted for only 0.15% of the total, mainly belonging to the families *Methanobacteriaceae*, *Methanocellaceae*, and *Methanosarcinaceae*.

In all samples at the phyla level, the top five phyla were *Firmicutes* (69.59%~85.20%), *Actinobacteria* (9.97%~25.48%), *Proteobacteria* (0.33%~3.69%), *Planctomycetota* (0.47%~1.71%), *Chloroflexi* (0.45%~1.61%) and other phyla (0.23%~0.70%). At the family level, the top five families were *Bacillaceae* (62.70%~81.70%), *Pseudonocardiaceae* (6.54%~16.39%), *Thermomonosporaceae* (1.52%~4.80%), *Clostridiaceae* (0.74%~3.16%), *Isosphaeraceae* (0.46%~1.70%), and other families (5.14%~10.35%) (Fig. 3). Compared with the oxygen-depleted treatments, the addition of oxygen increased the relative abundance of a few species, such as *Bacillaceae*, and decreased the relative abundance of most species. Oxygen was harmful to most species. However, *Bacillaceae* had the highest relative abundance in all treatments, and this increase in relative abundance may not indicate that *Bacillaceae* is more suitable for growth under aerobic conditions.

We calculated alpha diversity based on OTU level and analyzed the richness and diversity of soil microbial community using chao and sobs indices for richness and shannon and Simpson indices for diversity. These four indices showed similar changes under all treatments, and they could be ranked as N24 > N12 > O12 >O24. The richness and diversity of soil microbial community in oxygen-limited treatments (O24, O12) were significantly lower than those in oxygen-depleted treatments (N24, N12), but the short-term ventilation changes (O12, N12) did not cause large fluctuations as soil properties did (Fig. S2).

According to the Venn diagram, 712 OTUs were shared by all four treatments, accounting for 49.89% of the total. The N24 group had 87 OTUs that were unique to it, the N12 group had 44 OTUs that were unique to it, the O12 group had 16 OTUs that were unique to it, and the O24 group had 11 OTUs that were unique to it. The number of unique OTUs decreased with increasing oxygen pressure, suggesting that fewer OTUs in the microbial community could adapt to the oxygen environment, and thus the richness and diversity of the microbial community also declined (Fig. S3A). The PC1 axis obtained by PCoA analysis explained 84.07% of the variation. The four treatments were clearly divided into oxygen-depleted groups and oxygen-limited groups on PC1 axis, which also indicated that stable oxygenation could significantly

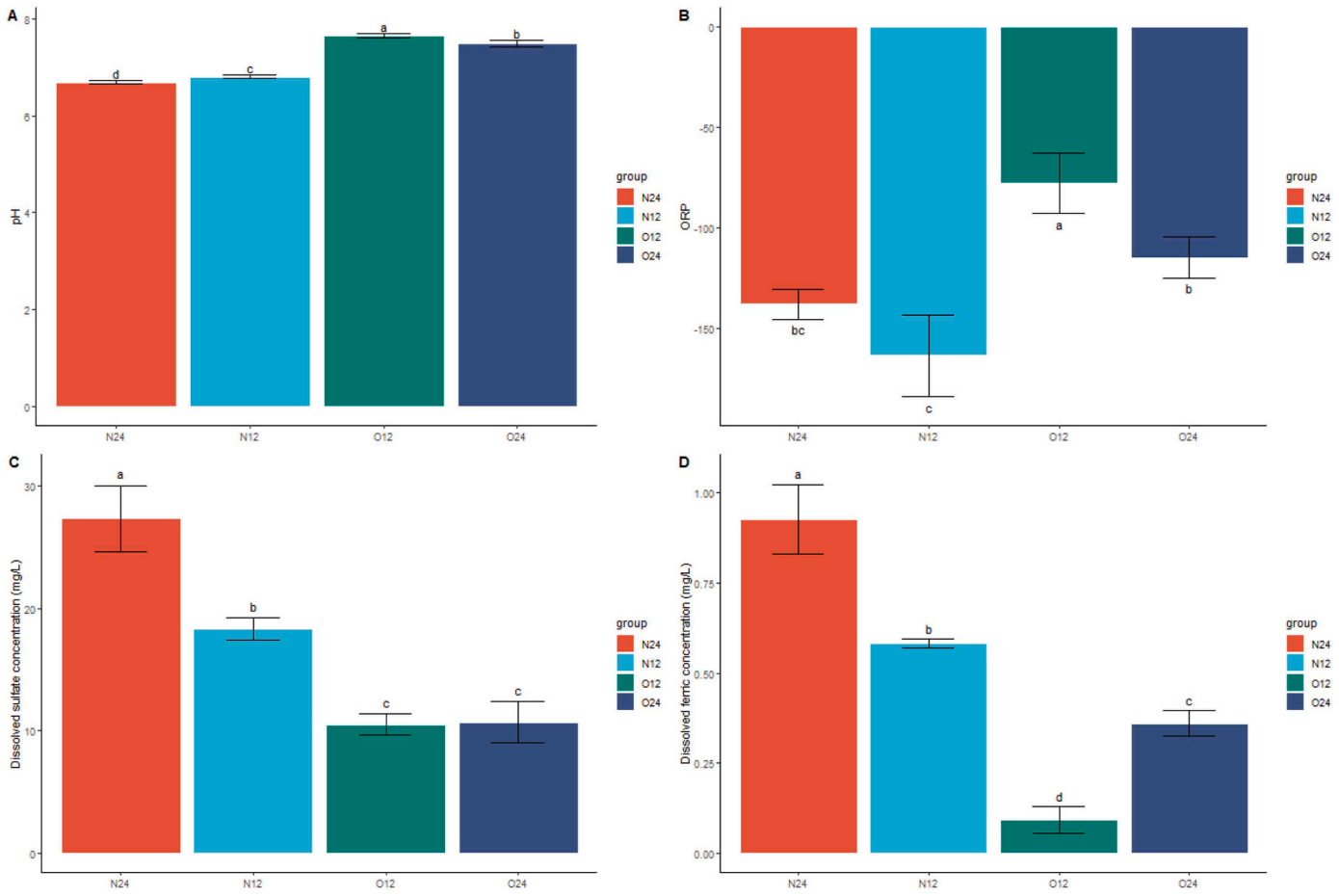


Fig. 2. Effect of nitrogn and/or oxygen amendment on soil properties. (A) pH, (B) ORP, (C) Dissolved sulfate, (D) Dissolved ferric. The lowercase letters and corresponding error bars indicate significant differences ($p < 0.05$). Data are shown as the mean ± standard deviation.

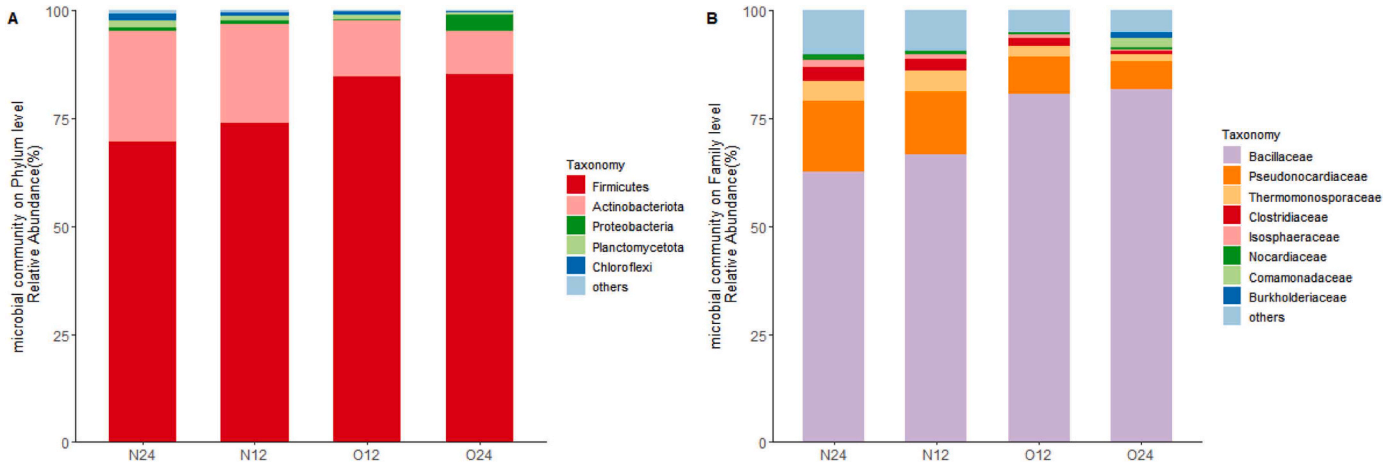


Fig. 3. Microbial community abundance of *16S rRNA* sequencing at different levels. (A) The percentage of community abundance at the phylum level, (B) the percentage of community abundance at the family level.

change the microbial community composition, but the short-term oxygen change had limited effect on the community composition, or the community composition change was completed within 12 h of ventilation interval (Fig. S3B).

3.3. *hgcA* sequencing community composition of the paddy soil

The community composition of Hg methylators at the phylum and family levels was shown using amplicon sequencing of *hgcA*. We identified a broad taxonomic diversity of Hg methylators in different treatments. After degeneracy and screening, 449 OTUs were obtained, of which 226 (50.33%) could not be assigned to the genus. The *hgcA* genes were assigned to 11 distinct bacterial and archaeal phyla, but could only be further assigned to 10 families. Of the 449 OTUs, 23 OTUs (2.22%) remained unclassified at the phylum level and 226 OTUs (17.47%) remained unclassified at the family level.

The ten most abundant phyla found in all groups were *Firmicutes* (29.49%~41.34%), *Euryarchaeota* (22.51%~33.38%), *Proteobacteria* (22.53%~28.16%), Chloroflexi (2.21%~3.14%), *Chlorobi* (1.80%~2.70%), *Unclassified* (1.53%~2.56%), *Lentisphaerae* (1.62%~2.13%), *Elusimicrobia* (0.95%~1.71%), *Candidatus Aminicenantes* (0.63%~1.46%), *Spirochaetes* (0.75%~0.99%). The abundant families found in all groups were *Peptococcaceae* (24.62%~37.88%), *Methanocellaceae* (19.92%~28.42%), *Geobacteraceae* (18.86%~25.11%), *Unclassified* (15.06%~18.32%), *Clostridiaceae* (1.61%~2.54%), *Ruminococcaceae* (0.82%~1.98%), *Methanosarcinaceae* (0.62%~1.31%), *Methanoregulaceae* (0.43%~0.99%), *Syntrophaceae* (0.46%~0.76%), *Spirochaetaceae* (0.22%~0.30%), *Desulfovibrionaceae* (0.13%~0.21%), *Desulfuromonadaceae* (0.01%~0.09%) (Fig. 4).

Peptococcaceae, *Clostridiaceae*, and *Ruminococcaceae* are known to contain some genera that act as fermenters, while *Methanocellaceae* are putative methanogens and *Geobacteraceae* are putative sulfate or iron reducers. The results also showed that the addition of oxygen influenced the microbial community structure of Hg methylators. The oxygen-limited groups increased the relative abundance of most species compared with the oxygen-depleted groups. For example, *Peptococcaceae* surged from 24.65% in N24 to 33.24% in O24. Other species that increased were *Ruminococcaceae*, *Methanoregulaceae*, and others. The oxygen-limited groups also reduced the relative abundance of a few species. For instance, *Methanocellaceae* and *Geacterobaceae* declined from 25.02% and 25.11% in N24 to 22.99% and 18.86% in O24, respectively.

We also analyzed the alpha diversity of methylation microbial community measured by *hgcA* at OTU level. Although by comparing the means, we still found that the richness and diversity of methylation microbial community in oxygen-limited groups were lower than those in oxygen-depleted groups, which was consistent with the sequencing results based on *16S rRNA*, but ANOVA analysis did not find significant differences among these groups (Fig. S4).

We used the Sampling RDA method to identify the key factors influencing the microbial community and the *hgcA* community. These two RDA axes explained 87.82% of the total variance in the *16S rRNA* sequencing community and 57.52% in the *hgcA* sequencing community. The RDA diagram displays how species data (points) and environmental factors (arrows) are related. The direction and length of an arrow indicate the correlation and contribution of an environmental factor to a sorting axis. ORP and pH had a positive correlation with RDA1, while sulfate and ferric had a negative correlation with RDA1. This suggests that environmental factors can be grouped into two categories based on their influence on species composition. The distance between samples represents species similarity. *16S rRNA* samples were separated into oxygen-limited groups and oxygen-depleted groups, while *hgcA* samples were mixed together, which was consistent with the alpha diversity results (Fig. S5).

3.4. The function of bacterial communities in paddy soil

KEGG pathway enrichment analysis was used to annotate the functional content of the bacterial community in flooded paddy soil under different ventilation, to examine how microbial community changes affect metabolic functional diversity. The most abundant KEGG pathways are those of metabolism (76.9–78.4%) under four ventilation conditions, followed by genetic information processing (5.5–6.7%), environmental information processing (5.2–6.3%), cellular processes (4.5–5.2%), human diseases (3.4–4.1%), and organismal systems (1.8–2.0%) (Fig. 5).

Cluster analysis results showed that the samples were divided into oxygen-depleted groups and oxygen-limited groups based on the ventilation status, and the KEGG pathways were classified into three categories: more expressed in oxygen-depleted groups, more expressed in oxygen-limited groups, and no obvious relationship with ventilation status. More than half of the KEGG pathways expressed more when oxygen was supplied. These included most metabolic pathways, such as energy metabolism, nucleic acid metabolism, carbohydrate metabolism, transport and catabolism, glycan biosynthesis and metabolism, lipid metabolism. These metabolic pathways involved both anabolism and catabolism, indicating that the introduction of oxygen promoted the activity and growth of microorganisms in paddy soil, because microbial metabolism refers to the various metabolic activities of microorganisms during their survival, including catabolism and anabolism. Catabolism refers to the oxidation, exothermic and enzymatic degradation of organic substances by microorganisms to break down complex large molecules into small molecules. Anabolism refers to the process of heat absorption and enzyme synthesis by microorganisms using nutrients and energy released in catabolism to make microorganisms grow and proliferate.

4. Discuss

4.1. MeHg formation in paddy soil under different ventilation conditions

The experiment designed four treatments, including two full-pressure groups (O24, N24), to directly investigate whether air amendment would increase MeHg formation compared to nitrogen amendment. As far as we know, most of the current soil culture experiments are conducted under sealed conditions to study MeHg formation

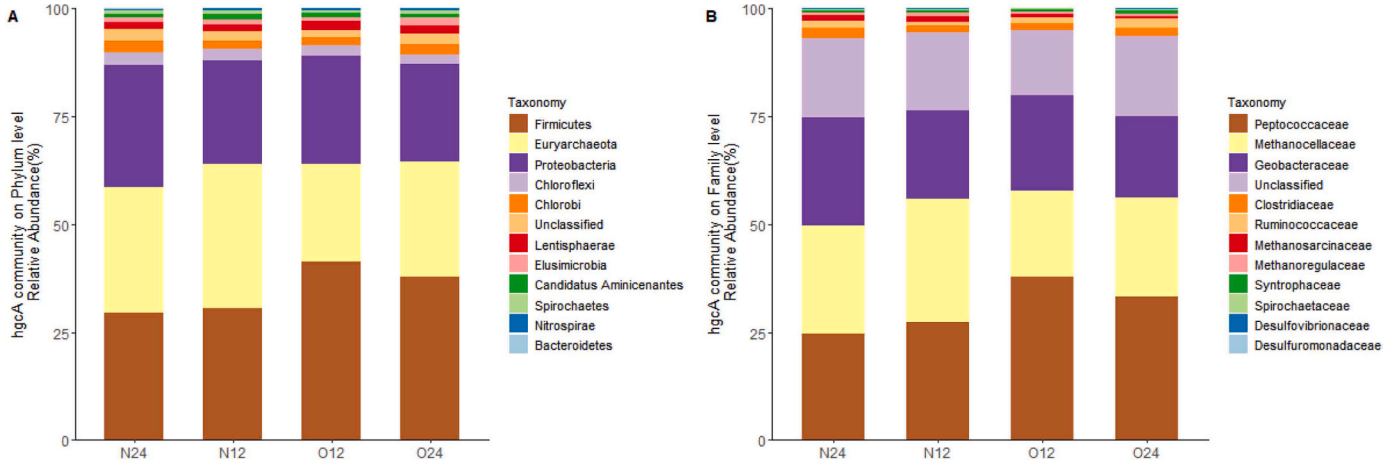


Fig. 4. Microbial community abundance of *hgcA* sequencing at different levels. (A) The percentage of community abundance at the phylum level, (B) the percentage of community abundance at the family level.

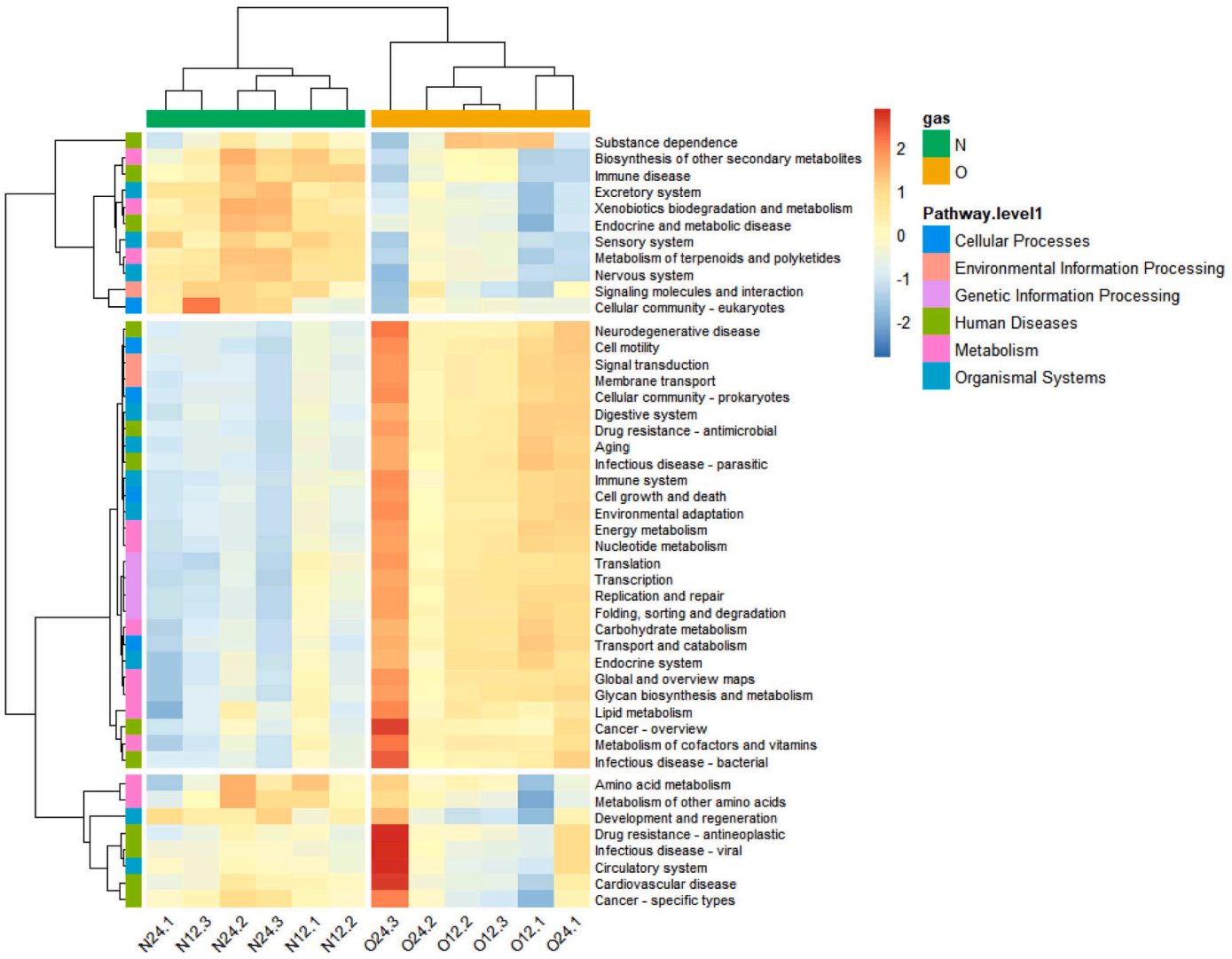


Fig. 5. Using the metagenomes estimated by PICRUSt, we analyzed functional genes encoding proteins associated with microbial turnover to investigate changes in Hg cycling potential among various ventilatory conditions.

at different Eh, pH, or sulfate concentrations (Fuhrmann et al., 2021; C. Liu et al., 2021; Millera Ferriz et al., 2021; Pierce et al., 2022; Wang et al., 2021), and the soil culture is not exposed to external air. Except for (Zhu et al., 2018), who incubated river bottom sediments with through-air treatment, resulting in maximum MeHg production with moderate air exposure. However, Zhu et al. did not measure DO simultaneously (Zhu et al., 2018), so they could not directly determine whether their soil oxygen concentration was excessive. We hypothesized that in their experiment, moderate air exposure introduced oxygen into the soil, but did not change its anaerobic state. This anaerobic soil environment with oxygen input facilitated MeHg formation. This hypothesis was confirmed in our experiment, because oxygen was supplied to the paddy soil in the experiment, but the concentration of dissolved oxygen in all the control groups was still only 0.01 ppm, indicating that the rate of oxygen access was slower than the rate of oxygen consumption. The results in our experiment showed that low oxygen input significantly increased the concentrations of MeHg in the soil compared to the pure nitrogen control (Fig. 1). Thus, we could infer that oxygen-limited anaerobic paddy soil have favor to MeHg formation compared to oxygen-depleted soil. Zhu et al.'s experiment increased sulfate concentration with increasing time of exposure to synthetic air due to the degree of oxidation of the system increasing. In contrast, sulfate concentration in this experiment decreased with increasing time of exposure to synthetic air (Fig. 2). This might be because their experiment had more oxygen input, enough to oxidize sulfides (Zhu et al., 2018).

The experiment also included two half-pressure groups (N12, O12), to investigate the effect of transient fluctuation of the aeration on MeHg formation. Many studies have observed that changing environments favor MeHg formation, such as higher MeHg in the water level fluctuation group than in the static water group in the field experiments, as water level fluctuations facilitate the diffusion of oxygen into the soil (Eckley et al., 2017; Xiang et al., 2018), plowed rice soil is more conducive to Hg methylation than unplowed rice soil (Sun et al., 2019). Our results showed that the Hg variables results tended to depend on the final gas condition (Fig. 1). This might be because MeHg synthesis could reach its maximum value within 12 h (Pierce et al., 2022). But soil property variables results tended to be divided into four groups, indicating that transient gas condition alter had different effects on soil properties. Particularly interesting was that soil property extremes generally occurred in half-pressure groups, indicating that gas condition transition had a greater impact on soil properties (Fig. 2).

We analyzed the correlation between Hg variables (THg, MeHg, and %MeHg) and soil properties (pH, ORP, sulfate, ferric), and found that both MeHg concentration and %MeHg were positively correlated with pH and ORP, and negatively correlated with sulfate and ferric. The

difference was that the correlation of %MeHg was statistically significant, while that of MeHg concentration was not. The correlation of THg concentration with the above four factors was different from that of MeHg, i.e., it was significantly negatively correlated with pH, ORP, sulfate, and positively correlated with ferric (Fig. S1). These results also emphasize that the changing environment has important effects on MeHg formation (W. Tang et al., 2020). Correlation analysis found that %MeHg was better correlated with soil properties than MeHg concentration (Fig. S1), indicating that%MeHg was more suitable as an evaluation criterion for MeHg formation (Yu et al., 2022). And %MeHg was highest in O12 group, reminding us again that MeHg formation under fluctuating conditions is a problem worth paying attention to.

4.2. Different ventilation conditions significantly altered the 16S rRNA sequencing community but had little effect on the hgcA sequencing community

Bacterial community characteristics are often represented by the relative abundance and diversity of species. The experimental operation divided the sample into two groups, oxygen-depleted group (N24, N12) and oxygen-limited group (O24, O12). We expect that the microbial community composition of these two groups will be very different. The major cause for this significant difference in bacterial communities is that oxygen acted as a filter for microorganisms in the environment, which could be explained by two reasons. First, microorganisms can be classified according to their tolerance to oxygen into anaerobes, aerobes, or facultative bacteria (Lin et al., 2021; Ma et al., 2019; Peterson et al., 2020). When oxygen entered the soil, anaerobes that were most sensitive to oxygen decreased significantly. This was evident from bacterial composition bar (Fig. 3) and Venn diagram (Fig. S4), where both species abundance and unique species numbers in oxygen-limited group were much lower than those in oxygen-depleted group. Second, oxygen significantly changed soil physicochemical properties and had a decisive impact on many substances in soil, such as replenishing electron acceptors to the soil (Bravo and Cosio, 2020; Pierce et al., 2022), oxidizing recalcitrant organic matter (C. Liu et al., 2021), and then serving as nutrient sources for some microorganisms.

After exposure to a small amount of oxygen, the species richness and abundance of the *16S rRNA* community decreased significantly. This result can be visually demonstrated by alpha diversity and venn diagram. Oxygen was harmful to most species, except *Bacillaceae*. However, *Bacillaceae* had the highest relative abundance in all treatments, and this increase in relative abundance may not indicate that *Bacillaceae* is more suitable for growth under aerobic conditions. The *hgcA* community was less affected by low oxygen input than the *16S rRNA* community. Alpha diversity showed no significant difference among groups, although numerically it was still lower in oxygen-limited groups than in oxygen-depleted groups (Fig. S4). Species abundance and unique species numbers also did not change significantly, but most species sequenced by *hgcA* increased their relative abundance in air amendment group (Fig. 4), such as *Peptococcaceae*, *Clostridiaceae*, and *Ruminococcaceae*. This might be because *hgcA* sequencing community were not strict anaerobes or could use oxygen so they were not so sensitive to low oxygen diffusion (Capo et al., 2020; Gionfriddo et al., 2020; Jones et al., 2020). We performed redundancy analysis (RDA) to figure out how changes in soil properties influenced *16S rRNA* community and *hgcA* community composition at OTU level. We found that soil properties had consistent effects on both communities: sulfate and ferric iron were negatively correlated with RDA1 axis, while pH and ORP were positively correlated with RDA1 axis (Fig. S5). This indicates that the change in soil properties affected the 16s and *hgcA* communities in the same way. By simultaneously measuring the *16S rRNA* community and the *hgcA* community, we can draw a surprising conclusion: low oxygen input had a great impact on *16S rRNA* sequencing community composition, but less impact on *hgcA* sequencing community composition. This contradicts our traditional impression of Hg methylators, because we usually describe them as anaerobes (Malcolm et al., 2010), so oxygen amendment should reduce *hgcA* community instead, but experimental results were opposite.

4.3. Metabolic pathway clustering reveals oxygen-induced enhancement of microbial activity in paddy soil

The presence of oxygen may influence the functions and processes of the paddy soil ecosystem by shaping distinct bacterial communities with different functional profiles (Li et al., 2023), among the oxygen-depleted and oxygen-limited groups. The oxygen-limited group showed higher predicted abundances of most KEGG pathways than the oxygen-depleted group, especially those related to metabolism (Fig. 5). We focused on the metabolic pathways because they reflect the microbial activity and Hg methylation potential in paddy soil (Großkopf et al., 2016; McDaniel et al., 2020; Wang et al., 2023). Hg methylation is known to be associated with sulfate reduction metabolic, iron reduction metabolic, and methane production metabolic, but its links with other metabolic pathways are unclear. Hg methylation involves energy metabolism, which affects the expression of *hgcA/hgcB* genes that encode Hg methyltransferase. For instance, *Desulfovibrio dechloroacetivorans* BerOc1 strain switches from sulfate reduction to respiratory iron metabolism when sulfate and acetyl chloride are absent, resulting in lower *hgcA/hgcB* gene expression levels (Goñi-Urriza et al., 2015). Moreover, cell metabolism provides precursor molecules and ATP for gene expression and regulates transcription factors and signaling pathways that modulate gene expression patterns (Goñi-Urriza et al., 2015). Hg methylation also depends on transport and degradation metabolism, which involves various substances entering and leaving cells and being degraded into energy or other molecules (Regnell et al., 2019). This may alter the intracellular concentrations and availability of Hg, cobalamin, sulfide etc., thereby influencing *hgcA* gene expression levels and activity. Therefore these metabolic pathways increase may be one of reasons for higher MeHg and %MeHg in oxygen-limited groups.

However, due to limited research, we cannot infer the direct relationship between other metabolic pathways and MeHg production, such as nucleic acid metabolism, carbohydrate metabolism, glycan biosynthesis and metabolism, and lipid metabolism, but these metabolic pathways increase represent the overall activity of the community. Previous studies have also found that MeHg production has a strong positive correlation with community activity increase when *hgcA* gene expression remains unchanged (Luo et al., 2023; Yu et al., 2018), which is consistent with our findings. Oxygen did not alter *hgcA* sequencing community structure, but enhanced overall community metabolic activity, and consequently increased MeHg concentration and potential.

5. Conclusion

We evaluated the effect of microbial community structure composed of *16 S rRNA* and *hgcA* gene sequences on MeHg production under different oxygen conditions. Adding oxygen increased the concentration of MeHg in the soil to 132.6 μg/kg, which is 1.77 times that of anaerobic soil. Ventilation conditions have an important effect on soil physical and chemical properties, but the peak appears in the half-pressure group, indicating that soil physical and chemical properties do not change linearly with ventilation time, and transient ventilation conditions can cause mutations in soil properties. Adding oxygen is not conducive to the *16 S rRNA* sequencing community, and species richness and abundance decrease because most anaerobic bacteria decrease in abundance and individual fermentation bacteria increase in abundance, such as *Bacillaceae*. In contrast, the *hgcA* sequencing community is almost unaffected by oxygen, and the abundance of most mercury methylating bacteria even increases, such as *Peptococcaceae*, *Ruminococcaceae*, *Methanoregulaceae* and others. This indicates that oxygen does not limit these mercury methylating bacteria. Adding oxygen also significantly increased the metabolic activity of the microbial community and

increased the abundance of more than half of the KEGG pathways, especially metabolic pathways related to methylation, including sulfate reduction metabolic, iron reduction metabolic, and methane production metabolic. For semi-open environments like paddy fields with half-dry and half-wet water level fluctuations, insufficient oxygen environment may be more likely to occur. This may also be the reason for the increase in MeHg concentration during dry-wet alternation (Q. Liu et al., 2021). Some studies expect to prevent methylation by adding oxygen (Shi et al., 2018), but excessive oxygen needs to be considered; otherwise, it may cause an increase in MeHg. Future research needs to explore the differences in microbial communities in hypoxic rice soils, especially their functions, to understand what drives MeHg production.

CRediT authorship contribution statement

Zihao Li: Formal analysis, Investigation, Methodology, Software, Visualization, Writing original draft, Writing-review and editing. **Tantan Wang**: Methodology, Writing-review. **Xu Yang**: Methodology. **Xin Wen**: Methodology. **Wenhao Chen**: Conceptualization, Data curation. **Zhigang Yu**: Project administration, Resources, Supervision. **Chang Zhang**: Conceptualization, Funding acquisition, Project administration, Resources, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2023.115585.

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